

Entropy-stable and physical-constraint-preserving DGSEM

for general-relativistic hydrodynamics

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Special-relativistic hydrodynamics: the conservation law

In flat Minkowski spacetime ($c = 1$): conservation of mass, momentum, and energy.

$$\boxed{\partial_t U + \sum_a \partial_{x_a} F^a(U) = 0}, \quad U = \begin{pmatrix} D \\ m_b \\ E \end{pmatrix}, \quad F^a(U) = \begin{pmatrix} Dv_a \\ m_b v_a + p\delta_{ab} \\ m_a \end{pmatrix}.$$

Conservative variables

Gamma-law gas, $1 < \Gamma \leq 2$

$$D = \rho L, \quad m_b = \rho h L^2 v_b, \quad E = \rho h L^2 - p.$$

$$L = (1 - |v|^2)^{-1/2}, \quad h = 1 + \frac{\Gamma p}{(\Gamma - 1)\rho}.$$

ρ : rest-mass density; p : pressure; v : fluid velocity; h : specific enthalpy.

Can we retain this familiar conservation-law structure in curved spacetime?

Martí-Müller, Numerical Hydrodynamics in Special Relativity, Living Rev. Relativ. 6, 7 (2003).

Two mathematical structures must survive discretization

Entropy inequality

$$\eta(U) = -\frac{D(\log p - \Gamma \log \rho)}{\Gamma - 1},$$

$$\partial_t \eta + \partial_a (v_a \eta) \leq 0.$$

Nonlinear stability across shocks; equality for smooth flow.

Physical admissibility

$$\rho > 0, \quad p > 0, \quad |v| < 1$$



$$D > 0, \quad q(U) := E - \sqrt{D^2 + |m|^2} > 0.$$

These states form an open convex cone $\mathcal{G}$.

Entropy stability alone does not enforce the physical constraints.

SRHD entropy: Duan–Tang (2019); admissible cone: Wu–Tang (2015).

GRHD is a balance law with geometry in flux and source

On a prescribed stationary spacetime, the perfect-fluid equations are

$$\nabla_\mu(\rho u^\mu) = 0, \quad \nabla_\mu T^{\mu\nu} = 0, \quad T^{\mu\nu} = \rho h u^\mu u^\nu + p g^{\mu\nu}.$$

With $L = (1 - \gamma_{ij} v^i v^j)^{-1/2}$, the 3 + 1 Valencia evolution law is:

$$\boxed{\partial_t(jU_V) + \partial_i(\alpha j F_V^i) = \alpha j S_V}, \quad j = \sqrt{\det \gamma}.$$

$$U_V = \begin{pmatrix} D \\ S_k \\ E \end{pmatrix}, \quad F_V^i = \begin{pmatrix} D \tilde{v}^i \\ S_k \tilde{v}^i + p \delta_k^i \\ E \tilde{v}^i + p v^i \end{pmatrix}, \quad \tilde{v}^i = v^i - \beta^i / \alpha, \quad S_k = \rho h L^2 \gamma_{k\ell} v^\ell.$$

α : lapse; β : shift; γ : spatial metric; u : four-velocity; g : spacetime metric.

$S_V(U_V; g, \partial g)$ is the geometric source; D and E retain their definitions from slide 2.

The metric changes admissibility and the flux–source entropy balance.

Banyuls et al. (1997); Font (2008). Stationary background; no Einstein evolution here.

A local frame recovers the metric-independent SRHD cone

Physical admissibility

$$D > 0, \quad q_\gamma(U_V) := E - \sqrt{D^2 + S^T \gamma^{-1} S} > 0.$$

Choose $\Theta^T \Theta = \gamma^{-1}$ and transform momentum and velocity:

$$\hat{m} = \Theta S, \quad \hat{v} = \Theta^{-T} v, \quad \hat{U} = (D, \hat{m}, E)^T.$$

$$|\hat{m}|^2 = S^T \gamma^{-1} S, \quad |\hat{v}|^2 = v^T \gamma v.$$

$$q_\gamma(U_V) = q(\hat{U}) = E - \sqrt{D^2 + |\hat{m}|^2} > 0.$$



one cone
at every point

The local-frame transformation gives the same admissible cone at every point.

Transformed-state / W-form framework: Wu, Physical Review D 95, 103001 (2017).

The local-frame equations: the W -form

For stationary geometry, set

$$\mathcal{L} = \text{diag}(1, \Theta, 1), \quad \hat{U} = \mathcal{L}U_V, \quad W = j\hat{U}.$$

Multiplying Valencia by $\mathcal{L}$ and applying the product rule gives

$$\boxed{\partial_t W + \partial_i H^i = \mathcal{S}_W}.$$

$$H^i = A_{ai}\hat{F}^a(\hat{U}) - C^i\hat{U}, \quad A_{ai} = \alpha j\Theta_{ai}, \quad C^i = j\beta^i,$$
$$\mathcal{S}_W = \alpha j\mathcal{L}\mathcal{S}_V + \underbrace{\alpha j(\partial_i \mathcal{L})F_V^i}_{\text{spatial frame variation}}.$$

Here $\hat{F}^a$ is the SRHD flux from slide 2, evaluated at $\hat{U}$.

Next, reduce this balance law while retaining the full physical volume measure.

Wu, Physical Review D 95, 103001 (2017). Full transformation derivation: B1.

Symmetry reduction: from $3 + 1$ to $2 + 1$

Suppress $x^\perp$: $\partial_\perp = 0$, $u^\perp = 0$; stationary block geometry, $I, a = 1, 2$.

$$w = \sqrt{\gamma_{\perp\perp}}, \quad j_2 = \sqrt{\det \gamma_{IJ}}, \quad \chi = w j_2,$$
$$\mathbb{W} = \chi \hat{U}, \quad \mathcal{A}_{aI} = \alpha \chi \Theta_{aI}, \quad C^I = \chi \beta^I.$$

$$\partial_t \mathbb{W} + \partial_I \underbrace{(\mathcal{A}_{aI} \hat{F}^a(\hat{U}) - C^I \hat{U})}_{\mathbb{H}^I: \text{SRHD fluxes with geometric coefficients}} = \mathbb{S}.$$

$\mathbb{S}$ includes both frame variation and the suppressed-direction source.

Planar flow: $w = 1$

Flat metric recovers $\partial_t U + \partial_I F^I(U) = 0$.

Axisymmetric flow: $w = r$

$\partial_t(rU) + \partial_r(rF^r) + \partial_z(rF^z) = (0, p, 0, 0)^T$.

A suppressed coordinate leaves its measure and pressure source in the PDE.

Our construction couples entropy stability, PCP, and OE

For the weighted GRHD system $\partial_t \mathbb{W} + \partial_l \mathbb{H}' = \mathbb{S}$:

Target	Construction	Result
Entropy stability	Complete geometric coefficients in both volume flux and discrete source	Semidiscrete entropy inequality
Physical constraints	Local-frame cone and weighted conservative scaling	Admissible averages and nodes
Oscillation control	Damp zero- χ -mean polynomial shells; retain the weighted anchor	Conservative OE, including a regular axis treatment

Foundations: Wu (2017); Zhang–Shu (2010); Peng–Sun–Wu (2025).

The key entropy construction: match the geometric terms

Entropy variables and potentials in the local frame:

$$V = \partial_{\hat{U}} \hat{\eta}, \quad \Phi = V^T \hat{U} - \hat{\eta} = D, \quad \psi_a = V^T \hat{F}^a - \hat{v}_a \hat{\eta} = D \hat{v}_a.$$

Volume flux: average the complete coefficients

$$\mathcal{H}_{pq}^{I,\text{EC}} = \{\{\mathcal{A}_{al}\}\}_{pq} \hat{F}_{pq}^{a,\text{EC}} - \{\{C^I\}\}_{pq} \hat{U}_{pq}^{\#}.$$

$[V]^T \hat{F}^{a,\text{EC}} = [\psi_a]$, $[V]^T \hat{U}^{\#} = [\Phi]$; $[\cdot]$: jump, $\{\{\cdot\}\}$: arithmetic mean.

Source: differentiate those coefficients with the same operator

$$V_p^T \mathcal{S}_p^h = \sum_I [\psi_{a,p} (\partial_I^h \mathcal{A}_{al})_p - \Phi_p (\partial_I^h C^I)_p].$$

These source terms cancel the geometric remainder from flux differencing.

SRHD EC flux: Duan–Tang (2019). ∂_I^h is the physical SBP derivative used on the next slide.

DGSEM: flux differencing, face dissipation, compatible source

Degree- N GLL nodal polynomials on affine elements; one coordinate line shown:

$$\dot{\mathbb{W}}_i + \underbrace{2\kappa \sum_m D_{im} \mathcal{H}_{im}^{\text{EC}}}_{\text{volume}} + \underbrace{\kappa \frac{B_{ii}}{\omega_i} (\hat{\mathbb{H}}_i - \mathbb{H}_i)}_{\text{faces}} = \underbrace{\mathbb{S}_i^h}_{\text{source}}.$$

In 2D, sum the two coordinate-line contributions. $\kappa = 2/\Delta x$, ω_i : GLL weights.

$$\underbrace{\mathbf{MD} + \mathbf{D}^T \mathbf{M}}_{\text{discrete integration by parts}} = \mathbf{B}, \quad \mathbf{M} = \text{diag}(\omega_i), \quad \partial^h = \kappa \mathbf{D}.$$

Local Lax–Friedrichs face flux with a causal speed

$$\hat{\mathbb{H}}_n = \frac{1}{2}(\mathbb{H}_n^- + \mathbb{H}_n^+) - \frac{a_f}{2}(\mathbb{W}^+ - \mathbb{W}^-), \quad a_f = |\beta \cdot n| + \alpha \sqrt{n^T \gamma^{-1} n}.$$

SBP supplies integration by parts; matching geometry supplies cancellation.

SBP flux differencing: Chen–Shu (2017); Gassner–Winters–Kopriva (2016).

Main result: a semidiscrete entropy inequality for GRHD

Theorem

On a conforming affine tensor-product mesh, assume admissible nodal states, $\chi > 0$, single-valued stationary face geometry, and periodic boundaries. The matched DGSEM with causal LLF faces satisfies

$$\boxed{\frac{d\mathcal{E}_h}{dt} \leq 0}, \quad \mathcal{E}_h = \sum_K \sum_p \mu_{K,p} \chi_p \hat{\eta}(\hat{U}_p).$$

Here $\mu_{K,p}$ are the mapped GLL quadrature weights.

Also conserved

The global weighted rest mass,
 $\sum_{K,p} \mu_{K,p} (\mathbb{W}_p)_D.$

Still to enforce

The assumption $\hat{U}_p \in \mathcal{G}$: an entropy bound does not guarantee it.

Next: preserve the admissible cone without changing conservative averages.

PCP: an admissible average anchors conservative nodal scaling

1. **Protect the average.** LF splitting and a source-aware sufficient timestep give

$$\Delta t(\lambda_{S,p} + \Lambda_{K,p}^{\text{face}}) < 1 \quad \forall p \implies \overline{\mathbb{W}}_K^{n+1} \in \mathcal{G}.$$

$\lambda_{S,p}$: source escape rate; $\Lambda_{K,p}^{\text{face}}$: face-speed / quadrature contribution; admissible input states and $\chi > 0$.

2. **Use the physical measure in the limiter.**

$$\hat{U}_{A,K} = \frac{\overline{\mathbb{W}}_K}{\bar{\chi}_K}, \quad \mathbb{W}_p^{\text{ref}} = \chi_p \hat{U}_{A,K}, \quad \overline{\mathbb{W}}_K^{\text{ref}} = \overline{\mathbb{W}}_K.$$

$$\boxed{\mathbb{W}_p^L = \mathbb{W}_p^{\text{ref}} + \theta_K(\mathbb{W}_p - \mathbb{W}_p^{\text{ref}})}, \quad 0 \leq \theta_K \leq 1.$$

Choose a common contraction to enter $\mathcal{G}$: $\overline{\mathbb{W}}_K^L = \overline{\mathbb{W}}_K, \quad \hat{U}_p^L \in \mathcal{G}.$

Zhang–Shu (2010); Wu–Tang (2015); Wu (2017). Production limiter adds density, cone and relative-margin safeguards; B7.

Conservative OE damps modes with zero weighted mean

Degree- r shells $Z_{r,K}$ from χ -weighted polynomial projections:

$$\hat{U}_h = \hat{U}_{A,K} + \sum_{r=1}^N Z_{r,K}, \quad \sum_p \mu_{K,p} \chi_p Z_{r,K,p} = 0.$$

$$\mathbb{W}_p^{\text{OE}} = \chi_p \left(\hat{U}_{A,K} + \sum_{r=1}^N \alpha_{r,K} Z_{r,K,p} \right), \quad \alpha_{r,K} = \exp \left[-s_{\text{OE}} \Delta t \sum_{m=0}^r \delta_m^{(K)} \right].$$

Exact average preservation

$$\overline{\mathbb{W}^{\text{OE}}}_K = \overline{\mathbb{W}}_K.$$

Directional damping rates $\delta_m^{(K)} \geq 0$

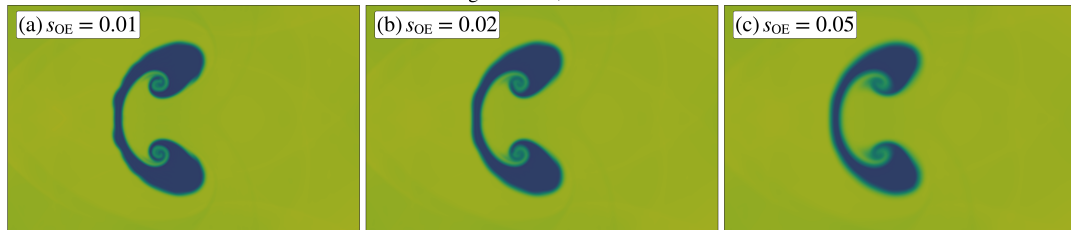
Derivative jumps, normalized on matching coordinate lines; causal speeds set the time scale.

The same weighted anchor supports both PCP and oscillation elimination.

OE framework: Peng–Sun–Wu (2025); GRHD PCP–OE: Cao–Peng–Wu (2025). Weighted hierarchy and sensor details: B8–B9.

The damping strength has a visible effect

Light bubble, $t = 450$



$s_{\text{OE}} = 0.01$

Finer visible structure

$s_{\text{OE}} = 0.02$

Default in these tests

$s_{\text{OE}} = 0.05$

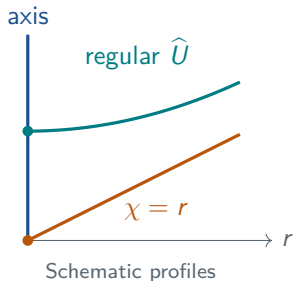
Stronger smoothing

The choice balances visible detail and regularization at this resolution.

Light shock–bubble, $t = 450$; same $\mathbb{Q}_2$ setup. An empirical choice, not a universal optimum.

Axis nodes: reconstruct the regular state from the weighted zero

For axisymmetric flow, $\chi = r$ and $\mathbb{W} = r\hat{U}$. A finite physical state therefore has $\mathbb{W}_\star = 0$ at $r = 0$.



At a simple zero of χ , use a derivative ratio:

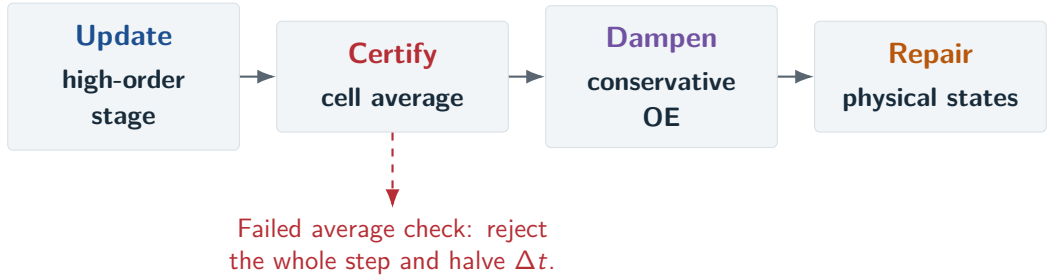
$$\hat{U}_\star \approx \frac{(D_I \mathbb{W})_\star}{(D_I \chi)_\star}.$$

Apply OE to the regular trace, then map back:

$$\mathbb{W}_\star^{\text{OE}} = 0, \quad \overline{\mathbb{W}}_K^{\text{OE}} = \overline{\mathbb{W}}_K.$$

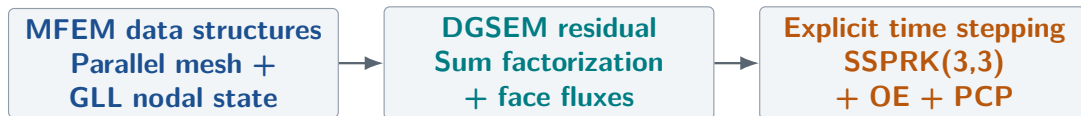
A separate regular-axis construction; the positive-weight entropy theorem does not automatically cover $\chi = 0$.

Every accepted stage follows the same order



The method separates entropy control, conservative damping, and admissibility.

DGSEM implementation with MFEM



MFEM infrastructure we use

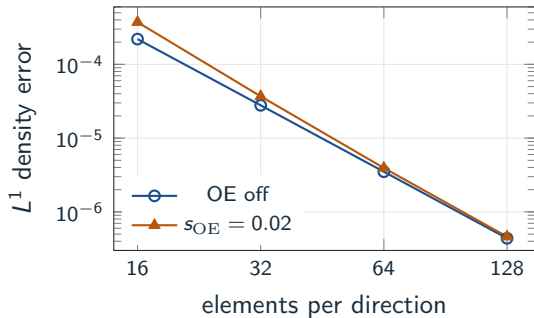
- ParMesh: mesh and element geometry.
- GLL DG_FECollection and ParFiniteElementSpace.
- ParGridFunction: nodal state and MPI face-neighbor exchange.

GRHD kernels we implement

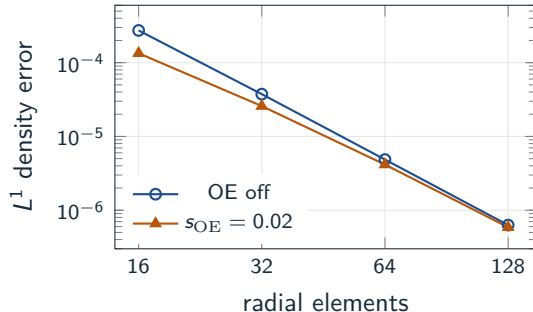
- Sum factorization: flux differencing along one-dimensional tensor lines.
- Cached stationary geometry; matched GRHD flux/source evaluation.
- Pointwise inverse mass; explicit stages with conservative OE and PCP limiting.

Smooth accuracy survives the stabilization

Flat spacetime: smooth wave

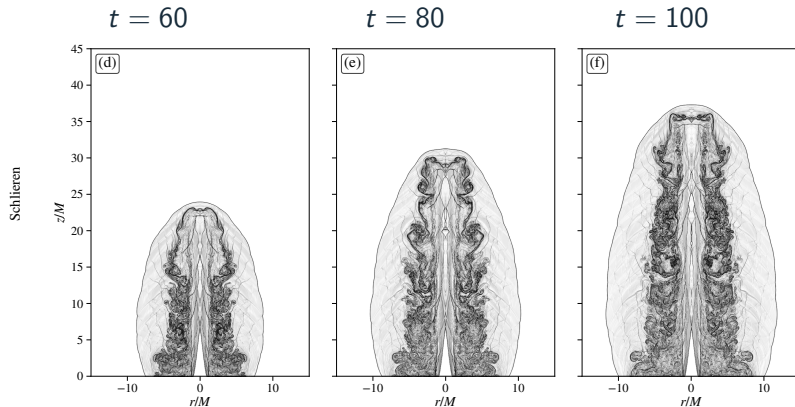


Schwarzschild: Michel accretion



Both tests approach third order for quadratic elements.

Axisymmetric jet: shocks and shear with a vanishing weight



Shocks

A propagating bow shock.

Shear

Cocoon and rolled-up structures.

Axis

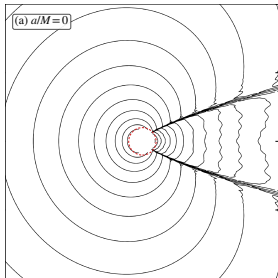
The weight vanishes at $r = 0$.

The combined method retains visible flow structure in this demanding test.

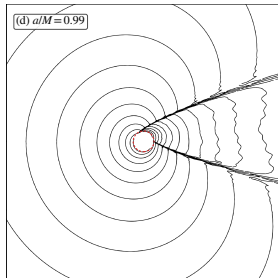
$\mathbb{Q}_2$, 360×1080 elements, PCP on, $s_{OE} = 0.02$; computed schlieren.

The same method also handles rotating black-hole backgrounds

Nonrotating: $a/M = 0$



Rapidly rotating: $a/M = 0.99$



Same spatial window in each panel: $-20 \leq x_{\text{KS}}/M, y_{\text{KS}}/M \leq 20$.

The large-scale shock cone is similar; spin mainly affects the near-hole flow.

Kerr Bondi–Hoyle, $t = 500M$, Kerr–Schild coordinates. Two of four spins shown; thin equatorial model, prescribed stationary metric.

Summary of the fully discrete scheme

Mathematical guarantees

Semidiscrete entropy stability under the stated assumptions.

Cell-average PCP under a sufficient timestep restriction; conservative nodal repair.

Numerical evidence

Near-third-order smooth accuracy for $\mathbb{Q}_2$.

Robust shock calculations with conservative OE and PCP on stationary backgrounds.

A fully discrete entropy inequality for RK–OE–PCP is not established here.

Curvilinear meshes, adaptive refinement, and evolving spacetimes remain future work.

Takeaways: GRHD structure in one compatible discretization

$$\partial_t(\chi \hat{U}) + \partial_l(\mathcal{A}_{al} \hat{F}^a - c^l \hat{U}) = \mathbb{S}$$

1. **Local-frame variables** recover SRHD fluxes and a metric-independent admissible cone.
2. **Matched geometric flux and source** yield a semidiscrete entropy-stable DGSEM.
3. **The same physical measure** χ makes PCP and OE preserve conservative averages, with a regular trace construction at the axis.

High-order accuracy and robust relativistic shock calculations on stationary backgrounds.

Preprint: [arXiv:2608.29229](https://arxiv.org/abs/2608.29229)

Questions?

From Valencia to the W -form

Start with the Valencia balance law

$$\partial_t(jU) + \partial_i(\alpha j F_V^i) = \alpha j \mathcal{S}_V, \quad j = \sqrt{\det \gamma}.$$

Choose a local orthonormal factor and its state transformation:

$$\Theta^T \Theta = \gamma^{-1}, \quad \mathcal{L} := \text{diag}(1, \Theta, 1).$$

Multiply Valencia by $\mathcal{L}$ and apply the product rule:

$$\mathcal{L} \partial_t(jU) = \partial_t(j \mathcal{L} U) - j(\partial_t \mathcal{L}) U,$$

$$\mathcal{L} \partial_i(\alpha j F_V^i) = \partial_i(\alpha j \mathcal{L} F_V^i) - \alpha j (\partial_i \mathcal{L}) F_V^i.$$

Define the transformed state, flux, and source

$$W := j \mathcal{L} U, \quad H^i := \alpha j \mathcal{L} F_V^i,$$

$$\mathcal{S}_W := \alpha j \mathcal{L} \mathcal{S}_V + \alpha j (\partial_i \mathcal{L}) F_V^i + j(\partial_t \mathcal{L}) U.$$

$$\partial_t W + \partial_i H^i = \mathcal{S}_W.$$

The key step is algebraic: transform Valencia before discretization.

Wu, Phys. Rev. D 95, 103001 (2017)

Technical backup

Concrete two-dimensional models

(active coordinates) | (suppressed coordinate), $w = \sqrt{\gamma_{\perp\perp}}$

Planar SRHD

$$(x, y) | z, \quad w = 1$$

Translation symmetry; baseline and accuracy tests.

Axisymmetric SRHD

$$(r, z) | \phi, \quad w = r$$

Rotational symmetry; relativistic jet.

Schwarzschild GRHD

$$(r, \theta) | \phi, \quad w = r \sin \theta$$

Rotational symmetry; Michel and Bondi–Hoyle accretion.

Equatorial Kerr GRHD

$$(r, \tilde{\phi}) | \theta, \quad w = r$$

Thin equatorial restriction—not a symmetry reduction.

One two-dimensional solver; the metric, weight, and source select the model.

Font–Ibáñez–Papadopoulos (1999); Lora–Clavijo–Guzmán (2013)

From the weighted PDE to the semidiscrete DGSEM

The symmetry-reduced W -form is the weighted balance law

$$\boxed{\partial_t \mathbb{W} + \partial_I \mathbb{H}^I = \mathbb{S}}, \quad \mathbb{W} = wW, \quad \mathbb{H}^I = wH^I, \quad \mathbb{S} = w\mathcal{S}_W + \mathbb{S}_\perp.$$

On an affine element, let $\kappa_I = 2/\Delta x_I$ and let D be the GLL-SBP differentiation matrix. The nodal scheme is

$$\begin{aligned} \dot{\mathbb{W}}_{ij} + \underbrace{2\kappa_1 \sum_{m=0}^N D_{im} \mathcal{H}_{(i,j),(m,j)}^{1,\text{EC}} + 2\kappa_2 \sum_{m=0}^N D_{jm} \mathcal{H}_{(i,j),(i,m)}^{2,\text{EC}}}_{\text{entropy-conservative volume flux differencing}} \\ + \underbrace{\frac{\kappa_1 B_{ii}}{\omega_i} \left(\widehat{\mathbb{H}}_{ij}^{1,\text{LLF}} - \mathbb{H}_{ij}^1 \right) + \frac{\kappa_2 B_{jj}}{\omega_j} \left(\widehat{\mathbb{H}}_{ij}^{2,\text{LLF}} - \mathbb{H}_{ij}^2 \right)}_{\text{LLF face corrections}} = \underbrace{\mathbb{S}_{ij}^h}_{\text{compatible source}}. \end{aligned}$$

The displayed scheme has three moving parts: EC volume flux, LLF face flux, and compatible source.

Unpack the three ingredients

1. Entropy-conservative volume flux

Average the *complete* coefficients $\mathcal{A}_{al} := wA_{al}$ and $\mathcal{C}^l := wC^l$:

$$\mathcal{H}_{pq}^{l,\text{EC}} = \{\{\mathcal{A}_{al}\}\}_{pq} \hat{F}_{pq}^{a,\text{EC}} - \{\{\mathcal{C}^l\}\}_{pq} \hat{U}_{pq}^\#.$$

2. LLF face flux

At an interface with spatial face normal n_I ,

$$\hat{\mathbb{H}}_n^{\text{LLF}} = \frac{1}{2}(\mathbb{H}_n^- + \mathbb{H}_n^+) - \frac{a_f}{2}(\mathbb{W}^+ - \mathbb{W}^-), \quad a_f = |\beta^I n_I| + \alpha \sqrt{\gamma^{IJ} n_I n_J}.$$

3. Compatible source discretization

Differentiate the same complete coefficients with the same SBP operators:

$$V_{ij}^T \mathbb{S}_{ij}^h = \sum_{l=1}^2 \kappa_l [\psi_{a,ij} (D_l \mathcal{A}_{al})_{ij} - \Phi_{ij} (D_l \mathcal{C}^l)_{ij}].$$

EC volume terms cancel the compatible source; LLF supplies entropy dissipation at faces.

Chen–Shu 2017; Gassner–Winters–Kopriva 2016; Duan–Tang SRHD EC flux

Causal LLF stabilization yields semidiscrete entropy stability

Lemma: causal LLF stabilization

For two admissible traces sharing the same prescribed face geometry, the causal LLF flux from the previous slide satisfies, with $[z] := z^+ - z^-$,

$$[V]^T \hat{\mathbb{H}}_n^{\text{LLF}} - [\Psi_{w,n}] \leq 0.$$

Theorem: semidiscrete DGSEM

On a conforming affine tensor-product mesh, assume admissible nodal states, $w > 0$, single-valued stationary face geometry, and periodic boundaries. Then

$$\frac{d\mathcal{E}_h}{dt} \leq 0,$$

global weighted rest mass is conserved.

The same geometry-only causal speed gives entropy stability here and PCP splitting next.

Lax–Friedrichs splitting keeps the cell average admissible

Lemma: Lax–Friedrichs PCP splitting

Wu–Tang 2015, Lemma 2.3; Wu 2017

At a coordinate face with spatial normal n_I , let

$$\lambda_n^{\text{geo}} = |\beta^I n_I| + \alpha \sqrt{\gamma^{IJ} n_I n_J}, \quad \begin{matrix} w > 0, \mathbb{W} \in \mathcal{G}, \\ a \geq \lambda_n^{\text{geo}} \end{matrix} \implies \boxed{\mathbb{W} \pm \frac{\mathbb{H}_n(\mathbb{W})}{a} \in \overline{\mathcal{G}}}.$$

Theorem: forward-Euler cell-average PCP

Zhang–Shu 2010; Qin–Shu–Yang 2016

Let

$$\Lambda_{K,\ell}^{\text{face}} := \frac{4}{\omega_{\ell_1} \omega_{\ell_2} |K|} \sum_{(f,q) \mapsto \ell} \omega_q J_{f,q} a_{f,q}.$$

With $\lambda_{S,\ell}$ the local source escape rate, admissible volume and face states, $w > 0$, and single-valued face geometry, the nodal condition

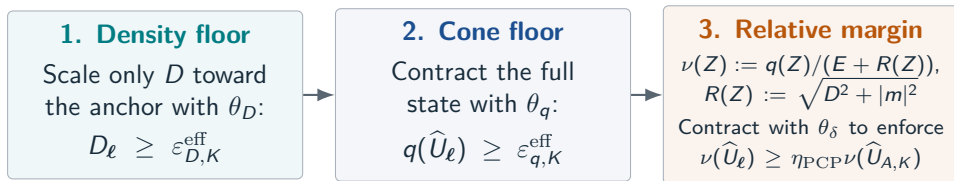
$$\boxed{\Delta t (\lambda_{S,\ell} + \Lambda_{K,\ell}^{\text{face}}) < 1 \quad \forall \ell} \implies \boxed{\overline{\mathbb{W}}_K^{n+1} \in \mathcal{G}}.$$

First certify the cell average; then conservative scaling restores nodal admissibility.

From an admissible average to admissible nodal states

Use the positive densitization $\chi_\ell = w_{\ell j_{2,\ell}}$ to build the local-state anchor and a conservative reference polynomial:

$$\hat{U}_{A,K} := \frac{\overline{W}_K}{\overline{\chi}_K} \in \mathcal{G}, \quad \mathbb{W}_\ell^{\text{ref}} := \chi_\ell \hat{U}_{A,K}, \quad \overline{\mathbb{W}}_\ell^{\text{ref}} = \overline{W}_K.$$



Each factor lies in $[0, 1]$ and is computed explicitly from a nodal minimum. Every stage is a contraction toward $\mathbb{W}^{\text{ref}}$, so

$$\overline{W}_K^L = \overline{W}_K, \quad \hat{U}_\ell^L \in \mathcal{G}_K^{\text{cert}} \quad \forall \ell.$$

Zhang–Shu 2010; Qin–Shu–Yang 2016; Wu 2017

Technical backup

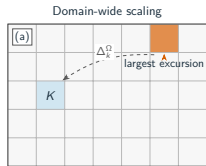
Directional OE: jumps normalized on matching coordinate lines

Design choice in this work: apply OE to the regular local state $\hat{U} = \mathbb{W}/\chi$.

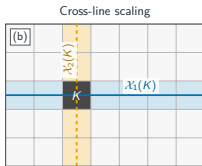
For derivative order m and direction l , use the RMS face jump $\mathcal{J}_{m,l,K}^{(k)}$ of $\partial_l^m \hat{U}^{(k)}$ and set

$$\sigma_{m,l}^{(K)} = \max_k \frac{(2m+1)(\Delta x_{K,l})^m}{2m! \Delta_{k,l,K}} \mathcal{J}_{m,l,K}^{(k)}.$$

$$\delta_m^{(K)} = \sum_{l=1}^2 \lambda_{K,l}^{\text{PCP}} \sigma_{m,l}^{(K)}, \quad \lambda_{K,l}^{\text{PCP}} \sim \frac{\text{causal speed}}{\Delta x_{K,l}/(2N+1)}.$$



one scale shared by all elements



direction-matched scales through K

$l = 1$: fixed- x^2 line, $l = 2$: fixed- x^1 line.
For the coordinate line $\chi_l(K)$ through K ,

$$\tilde{\Delta}_{k,l,K} := \max_{\substack{K' \in \chi_l(K) \\ \ell \in K'}} \left| \hat{U}_{K',\ell}^{(k)} - \overline{\hat{U}}_{\chi_l(K)}^{(k)} \right|,$$

$$\Delta_{k,l,K} = \max\{\tilde{\Delta}_{k,l,K}, 10^{-6} \Delta_k^{\Omega}\}.$$

Jump size says how much to damp; causal speed says how fast; the matching line supplies the scale.

A χ -weighted hierarchy makes OE conservative

Define the weighted anchor and hierarchical polynomial shells

$$\langle Z \rangle_{\chi,K} := \frac{\sum_{\ell} \mu_{K,\ell} \chi_{\ell} Z_{\ell}}{\sum_{\ell} \mu_{K,\ell} \chi_{\ell}}, \quad \hat{U}_{A,K} := \langle \hat{U}_h \rangle_{\chi,K}, \quad \hat{U}_h = \hat{U}_{A,K} + \sum_{r=1}^N \tilde{S}_K^r \hat{U}_h,$$

Here $P_{\chi,K}^r$ is the χ -weighted GLL projection, $\tilde{S}_K^r := P_{\chi,K}^r - P_{\chi,K}^{r-1}$, and every shell has zero χ -weighted mean.



$$\alpha_{r,K} = \exp \left[-s_{\text{OE}} \Delta t \sum_{m=0}^r \delta_m^{(K)} \right], \quad (\mathcal{O}_K^{\text{cons}} \mathbb{W}_h)_{\ell} = \chi_{\ell} \left[\hat{U}_{A,K} + \sum_{r=1}^N \alpha_{r,K} (\tilde{S}_K^r \hat{U}_h)_{\ell} \right].$$

$$\boxed{\sum_{\ell} \mu_{K,\ell} (\mathcal{O}_K^{\text{cons}} \mathbb{W}_h)_{\ell} = \sum_{\ell} \mu_{K,\ell} \mathbb{W}_{\ell}} \quad (\mathbb{Q}_2 : s_{\text{OE}} = 0.02 \text{ by default}).$$

OE damps hierarchical content, never the conservative cell average.

At a vanishing weight, store zero and reconstruct the finite trace

At an axis or polar node,

$$\chi_{\star} = 0,$$

$$\mathbb{W}_{\star} = 0,$$

$$\hat{U}_{\star} \approx \frac{(D_I \mathbb{W})_{\star}}{(D_I \chi)_{\star}}.$$

1 – Reconstruct

The endpoint derivative ratio gives a finite regular local state without dividing the stored zero by $\chi_{\star} = 0$.

2 – Apply OE

Use the same χ -weighted hierarchy, including the reconstructed trace, to damp the regular state.

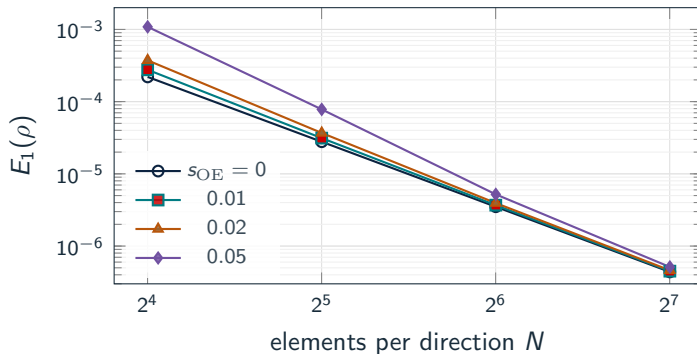
3 – Map back

Multiplication by χ restores the exact stored zero. Any raw-stage defect is conservatively redistributed along the normal GLL line.

$$\mathbb{W}_{\star}^{\text{OE}} = 0, \quad \sum_{\ell} \mu_{K,\ell} \mathbb{W}_{\ell}^{\text{OE}} = \sum_{\ell} \mu_{K,\ell} \mathbb{W}_{\ell}.$$

Stabilize a finite physical trace while preserving the exact weighted zero.

Smooth SRHD wave: OE retains the designed third order



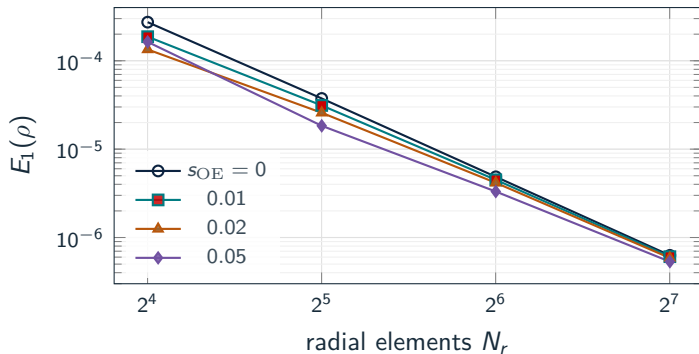
What to notice

All tested OE strengths recover approximately third-order convergence for $\mathbb{Q}_2$.

Finite-resolution cost

At 128^2 , $s_{OE} = 0.02$ increases density errors by only about 5–6% relative to OE off.

Michel accretion: curved-spacetime accuracy



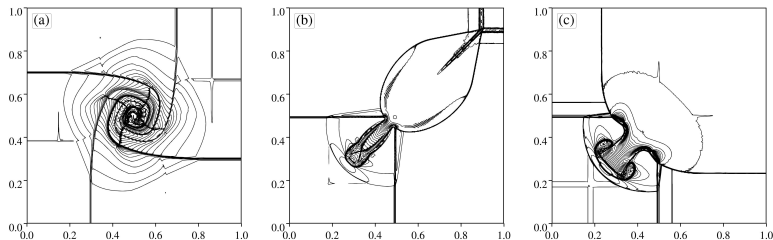
Steady exact solution

A smooth transonic Schwarzschild solution tests the full weighted GRHD geometry without shocks.

OE helps here

For this steady Michel test, $s_{OE} = 0.02$ gives lower $E_1(\rho)$ than OE off at every displayed resolution while retaining near-third-order convergence.

Two-dimensional Riemann problems: PCP under ultra-relativistic shocks



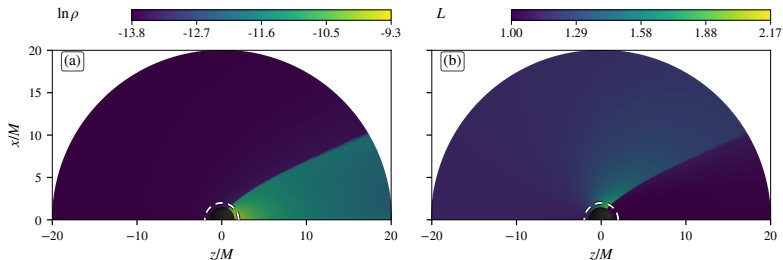
Three tests

Four-state interactions including ultra-relativistic states with velocity components near 0.99.

Production setup

$\mathbb{Q}_2$, 400×400 , CFL 0.8,
 $s_{\text{OE}} = 0.02$, PCP enabled.

Schwarzschild Bondi–Hoyle: a stable shock cone through $t = 500M$



M1a at $t = 500M$

Mach-5 asymptotic wind; the inner boundary lies inside the event horizon.

A well-defined downstream shock cone forms, with post-shock deceleration and near-horizon acceleration visible in the Lorentz factor.

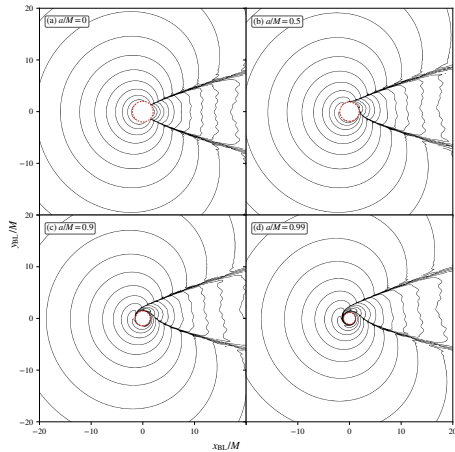
Numerical configurations

Common setup: $\mathbb{Q}_2$, CFL = 0.8, PCP on

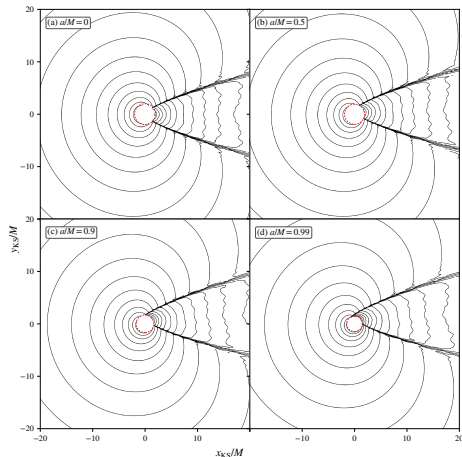
Test	w	Mesh	Final time
Smooth wave	1	$16^2\text{--}128^2$	0.1
Riemann	1	400^2	0.4
Shock–bubble	1	650×180	450/500
C2 jet	r	360×1080	100
Michel	$r \sin \theta$	$16 \times 4\text{--}128 \times 4$	$2M$
Schwarzschild M1a	$r \sin \theta$	256×128	$500M$
Kerr four spins	r	200×160	$500M$

$s_{\text{OE}} = 0.02$ for nonsmooth production runs; varied only in the smooth-wave, shock–bubble, and Michel studies.

Kerr Bondi–Hoyle in Boyer–Lindquist coordinates



Kerr Bondi–Hoyle: one method across spin $a/M = 0 \rightarrow 0.99$



Four spins, same setup

$\mathbb{Q}_2$, 200×160 , CFL 0.8,
 $s_{\text{OE}} = 0.02$.

Large-scale cone morphology is similar;
spin mainly distorts the near-hole flow.

Kerr–Schild view shown;
BL view in backup.